MIRA

Zhiyuan Li

Background

Background

MIRA: Medical Time Series Foundation Model for Real-World Health Data

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Background - Terminology

ECG (Electrocardiogram):
- Record the electrical activity of the heart

EEG (Electroencephalogram):
- Record the electrical activity of neurons in the cerebral cortex

↓

Understand the dynamic physiological states of patients over time

Background - Challenges

Irregular Sampling & Asynchrony:
- Medica data collected "as needed"
- *E.g. ECG tracked every ms, blood test every 12 hours*

Missing Values:
- Gaps in data, hard to maintain a continuous "picture" of the patient

Multi-Frequency Nature:
- Different timescales
- *E.g. Simultaneous high-frequency ECG and low-frequency vital signs*

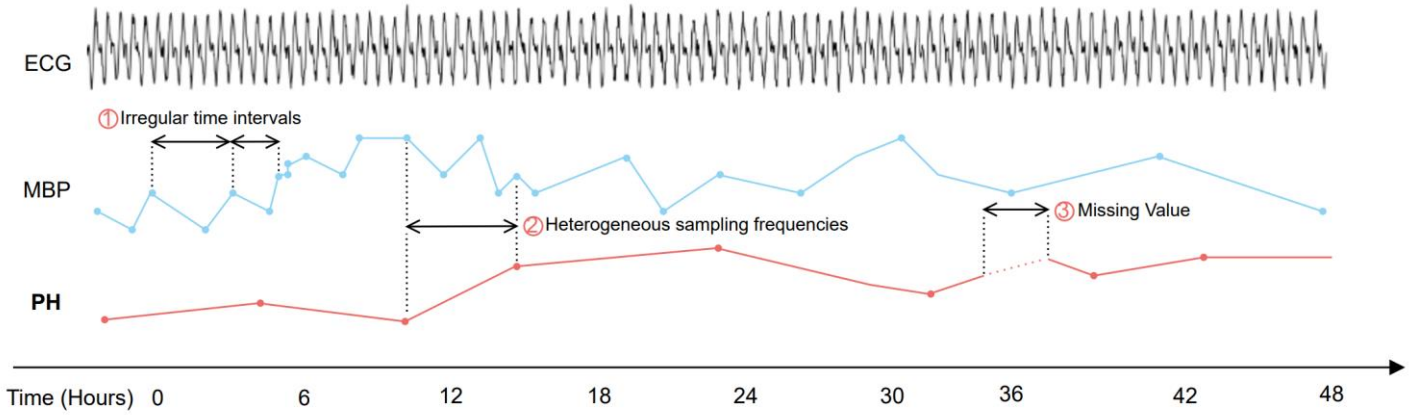


Figure 1: Medical time series exhibit ① irregular intervals, ② heterogeneous sampling rates, and ③ frequent missingness driven by clinical workflows.

Overview

How can we design a scalable and generalizable foundation model that captures the irregularity and multi-frequency nature of medical time series, while enabling robust transfer across diverse medical tasks?

MIRA - Medical Foundation Model For Irregularity Adaptation ***:***

- Foundation model specially for medical time series forecasting

Dataset:

- Extensive, diverse corpus of medical time series, ~ 454 billion observations for model pre-training

Methods

Irregular Medical Time Series Forecasting

Medical time series as **paired timestamps and observations**: $\{(t_i, x_i)\}_{i=1}^{N}$

- $t_i \in \mathbb{R}^+$: Timestamps, strictly increasing $(t_1 < \cdots < t_N)$
- $x_i \in \mathbb{R} \cup \{\mathrm{NaN}\}$: Observations, can be real or NaN (Missing)

2 Prevalent Irregular Patterns:

- ***Regular-grid Missing Values:*** $t_i = i\Delta t, x_i = \mathrm{NaN}$ for $i \in \mathcal{M}$
- The "grid" exists, but the "values" are missing

- **Irregular Sampling:** $\Delta t_i = t_{i+1} - t_i$
- Observations at the timestamps exist, spacing between them is uneven

Goal: $\hat{\mathbf{x}}_{L+1:L+H} = f_\theta(\{(x_i, t_i)\}_{i=1}^{L}; \{t_j\}_{j=L+1}^{L+H}) \in \mathbb{R}^H$

Irregular Medical Time Series Forecasting

Channel-Independent Modeling Strategy:

- MIRA treats each clinical variable *(E.g., Heart Rate, Temperature)* **independent sequence**
- Formally: $x^{(c)} = \{(t_1, x_1^{(c)}), (t_2, x_2^{(c)}), \ldots, (t_N, x_N^{(c)})\}$ for channel c
- Parameter Sharing: $\hat{x}^{(c)}_{L+1:L+H} = f_\theta\left(\{x_i^{(c)}, t_i\}_{i=1}^{L}; \{t_j\}_{j=L+1}^{L+H}\right)$

$$\Downarrow$$

$$[\hat{x}^{(c)}_{L+1}, \hat{x}^{(c)}_{L+2}, \ldots, \hat{x}^{(c)}_{L+H}] = f_\theta\left(\{x_i^{(c)}, t_i\}_{i=1}^{L};\ \{t_j\}_{j=L+1}^{L+H}\right)$$

MIRA Overview

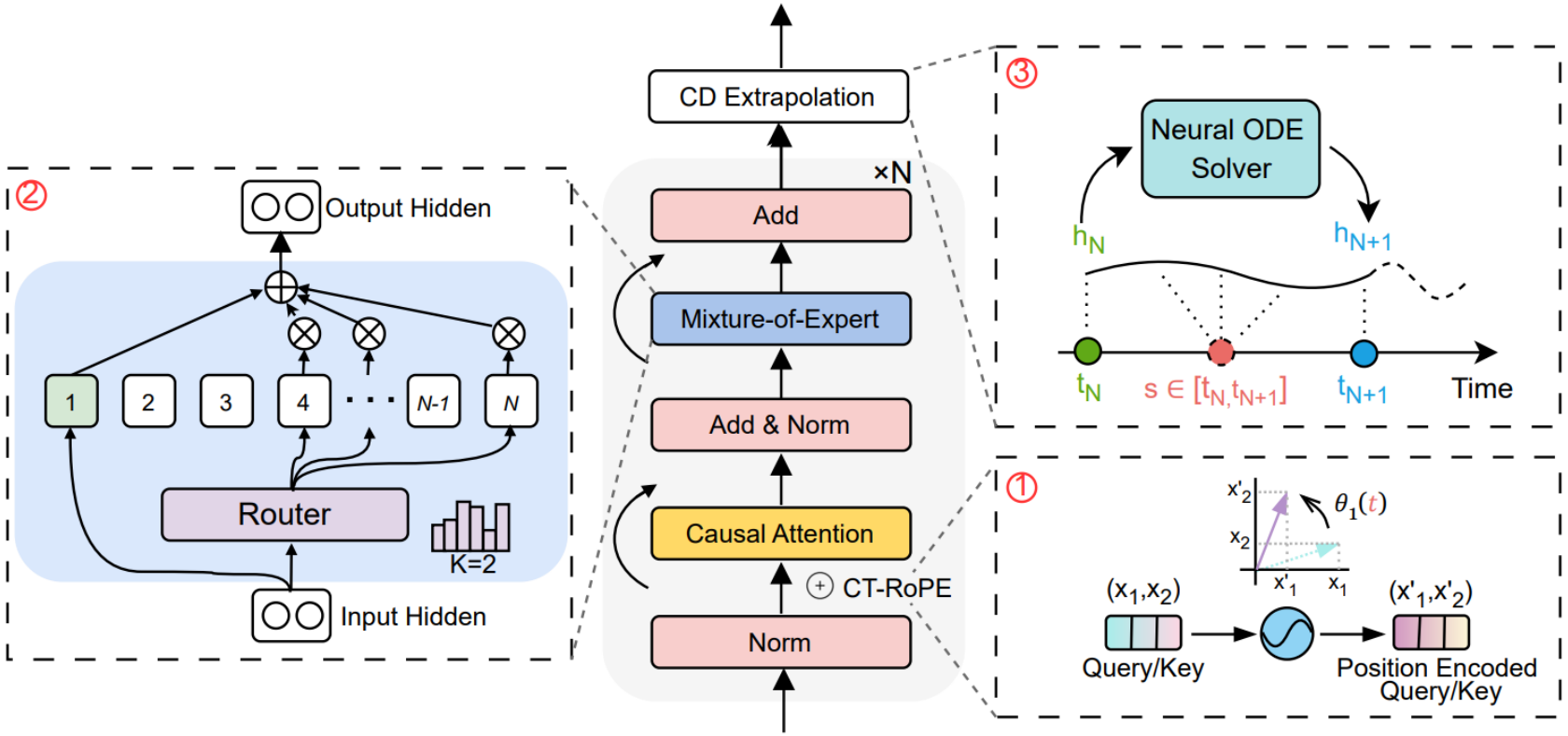


Figure 2: Architecture of **MIRA**. ① Takes irregular medical time series and timestamps as input, applying **CT-RoPE** for continuous temporal encoding. ② A **Sparse Temporal Mixture-of-Experts** layer routes tokens to specialized experts based on frequency. ③ A **Continuous Dynamics Extrapolation Block** evolves latent states toward arbitrary target timestamps for flexible time-aware forecasting.

Rotary Positional Encoding (RoPE)

Rotation is defined as **adding angles in polar coordinates**

- Preserves vector norm (no scaling)
- Two properties:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$R(\alpha)R(\beta) = R(\alpha + \beta)$$

$$R(\alpha)^\top R(\beta) = R(\beta - \alpha)$$

- Composition of rotations corresponds to angle addition
- Transpose represents the inverse rotation

RoPE

$$\mathbf{x} = \left[(x_0, x_1), (x_2, x_3), \ldots, (x_{d-2}, x_{d-1})\right]$$

High-dimensional vectors split into 2D subspaces

- Each pair -> own rotation + different angle
- *Why: Can be decomposed into sum of 2D rotations*

$$\theta_i = \omega_i \cdot m$$

$$\omega_i = 10000^{-2i/d}, \quad i = 0, 1, \ldots, d/2 - 1$$

- Each position m (integer) as a rotation angle
- Different frequencies for each i

RoPE

$$R_i(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \theta = \omega_i \cdot m$$

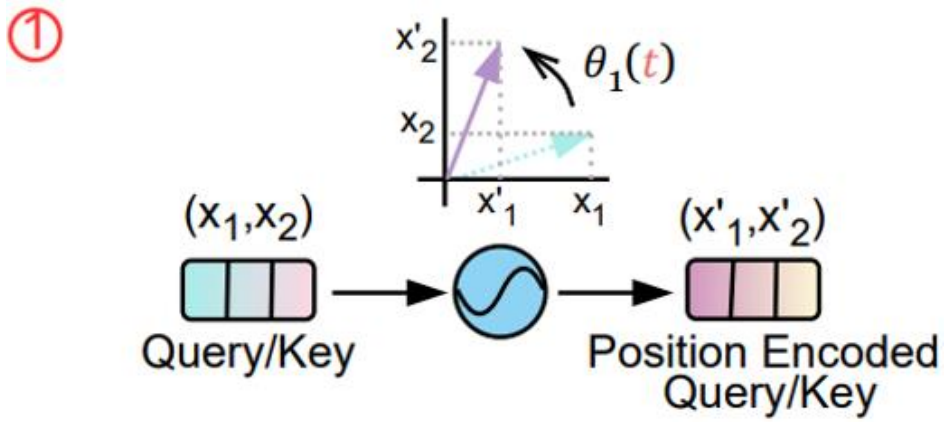
$$R(m) = \begin{bmatrix} R_0(\omega_0 m) & 0 & \cdots & 0 \\ 0 & R_1(\omega_1 m) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R_{d/2-1}(\omega_{d/2-1} m) \end{bmatrix}$$

$$\mathbf{q}_m \to R(m)\,\mathbf{q}, \quad \mathbf{k}_n \to R(n)\,\mathbf{k}$$

$$\mathbf{q}_m^\top \mathbf{k}_n = (R(m)\mathbf{q})^\top (R(n)\mathbf{k})$$

$$= \mathbf{q}^\top R(m)^\top R(n)\mathbf{k}$$

$$= \mathbf{q}^\top R(n-m)\mathbf{k}$$



The attention score depends only on **relative distance** $n - m$

Continuous-Time RoPE (CT-RoPE)

$$\theta_i(t) = \omega_i \cdot t, \quad t \in \mathbb{R}^+$$

$$(R(t_m)\mathbf{q})^\top (R(t_n)\mathbf{k}) = \mathbf{q}^\top R(t_n - t_m)\mathbf{k}$$

RoPE encodes **index distance** as phase difference

CT-RoPE generalizes to **real time difference** by replacing integer index with continuous timestamp

Attention becomes a function of temporal distance

CT-RoPE: Time Normalization

Given a strictly increasing timestamp sequence

$$\mathbf{t} = [t_1, t_2, \ldots, t_T],$$

we compute

$$t_{\min} = \min_i t_i, \qquad t_{\max} = \max_i t_i,$$

and rescale every timestamp into a canonical range that aligns with the sequence length:

$$\widehat{t}_i = \frac{t_i - t_{\min}}{t_{\max} - t_{\min} + \varepsilon}(T-1).$$

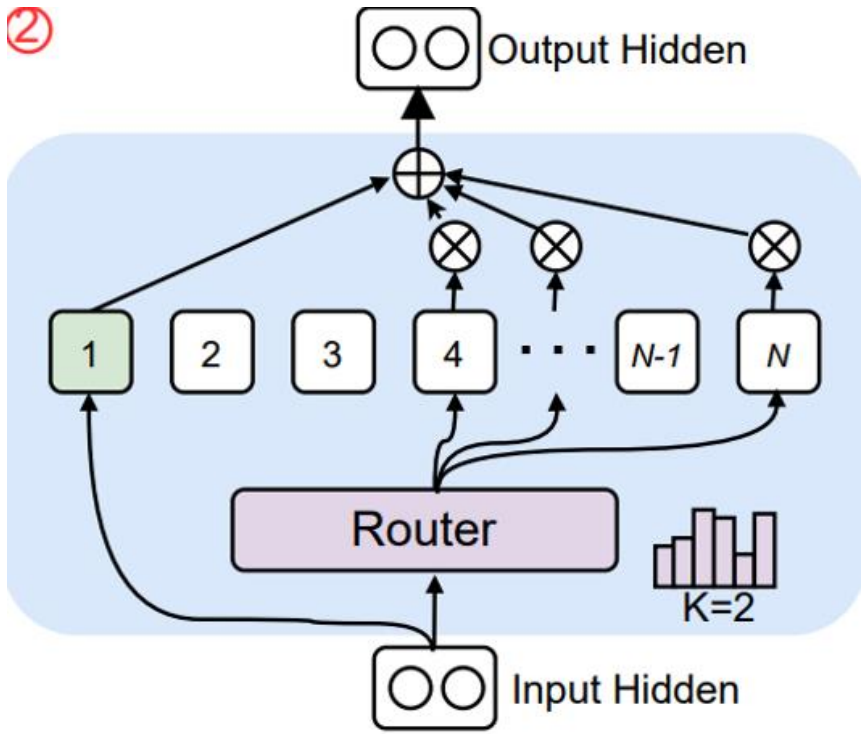
where, $(T-1)$ corresponds to the maximum index of a sequence of length T, ensuring that the normalized time domain matches the index range used in standard rotary encodings, while preserving the original temporal spacing in a continuous form. The small constant ε prevents numerical instability when the timestamp range is narrow [89, 90].

Frequency-Specific Mixture-of-Experts

Medical signals contain **multiple temporal frequencies**

Frequency-specific MoE:

- **What to Capture:** Local rate of change of the signal (fast vs. slow oscillations)
- **How:** Routes each token to specialized experts



Frequency-Specific Mixture-of-Experts

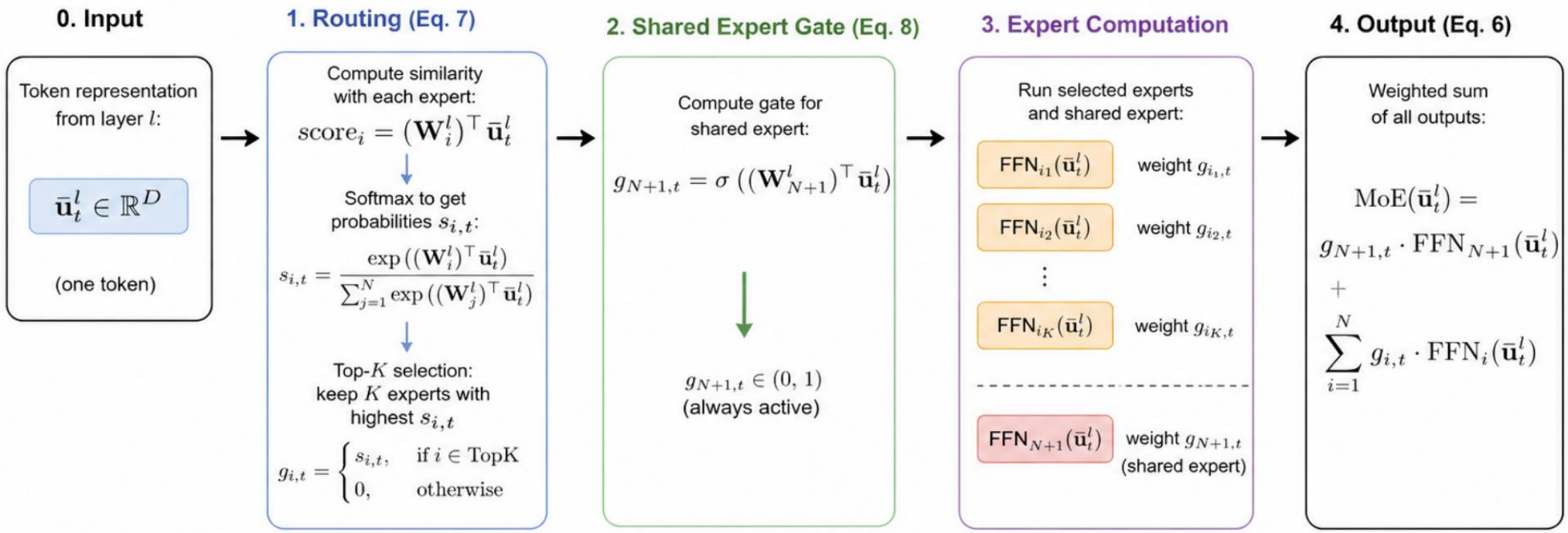
$$\mathrm{MoE}\left(\bar{\mathbf{u}}_t^l\right) = g_{N+1,t} \cdot \mathrm{FFN}_{N+1}\left(\bar{\mathbf{u}}_t^l\right) + \sum_{i=1}^{N} g_{i,t} \cdot \mathrm{FFN}_i\left(\bar{\mathbf{u}}_t^l\right)$$

$$g_{N+1,t} = \sigma\left(\left(\mathbf{W}_{N+1}^l\right)^\top \bar{\mathbf{u}}_t^l\right)$$

$$s_{i,t} = \frac{\exp\left((\mathbf{W}_i^l)^\top \bar{\mathbf{u}}_t^l\right)}{\sum_{j=1}^{N} \exp\left((\mathbf{W}_j^l)^\top \bar{\mathbf{u}}_t^l\right)}, \quad g_{i,t} = \begin{cases} s_{i,t}, & \text{if } s_{i,t} \in \mathrm{TopK}(\{s_{j,t}\}_{j=1}^{N}, K), \\ 0, & \text{otherwise,} \end{cases}$$

- $\boldsymbol{W}$ controls which experts to select
- g controls how much each selected expert contributes

Frequency-Specific Mixture-of-Experts



1. Routing for Non-Shared Experts (Eq. 7)

- $\mathbf{W}_i^l \in \mathbb{R}^D$: trainable gating vector for expert i (learns the "preference" of expert i)
- $s_{i,t}$: softmax score (similarity) between token and expert i
- $\text{TopK}(\cdot, K)$: select top-K experts with highest scores
- $g_{i,t}$: final routing weight after top-K selection (non-selected experts get weight 0)
- Only K experts are activated (sparse routing)

2. Shared Expert Gate (Eq. 8)

- $\mathbf{W}_{N+1}^l \in \mathbb{R}^D$: gating vector for shared expert
- $\sigma(\cdot)$: sigmoid function
- $g_{N+1,t} \in (0, 1)$: gate for shared expert
- Always active, not affected by Top-K selection
- Acts as a global residual pathway

3. Expert Computation & 4. Output (Eq. 6)

- Each FFN_i: a lightweight feedforward network (expert)
- Compute outputs of selected experts and shared expert
- Multiply by corresponding gates (weights)
- Sum them to get the final MoE output

Continuous Dynamics Extrapolation Block

Auto-regressive Architecture:
- Cannot incorporate the timestamp of the target token during inference
- Need to be **uniformly distributed**

Extrapolation:
- Predicting future values **beyond the known data**
- *E.g. Given blood pressure for past 10 hours, predict 12th hour value*

+

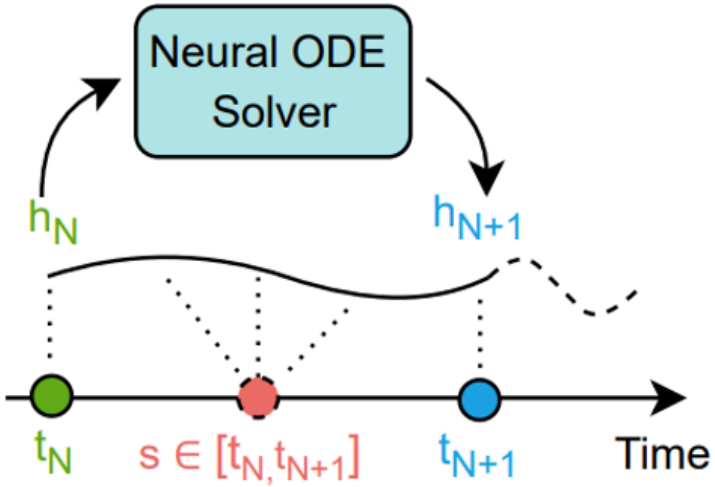
Neural ODE:
- Describe how a quantity changes continuously over time: $\frac{dh}{ds} = f(h)$
- Given initial state $h(t_N)$, integrate for future arbitrary t_{N+1}

Continuous Dynamics Extrapolation Block

$$\frac{dh(s)}{ds} = f(s - t_N, h(s); \theta_{ODE}), \quad h(t_N) = h_N, \quad s \in [t_N, t_{N+1}]$$

③

$$h(t_{N+1}) = h(t_N) + \int_{t_N}^{t_{N+1}} f(s - t_N, h(s); \theta_{ODE}) ds$$



The model starts from $h(t_N)$, and continuously "flows" through time using a learned dynamics function f, accumulating changes to reach $h(t_{N+1})$.

Continuous Dynamics Extrapolation Block

This integral is computed numerically using an ODE solver (i.e. the Dormand-Prince (RK45) method). Let the result of this numerical integration be denoted as $h'_{N+1} = h(t_{N+1})$. More detail is in Appendix D.

Implementation with Adaptive ODE Solvers We numerically approximate the solution to Equation 10 using adaptive step-size ODE solvers. Appropriate absolute and relative error tolerances (e.g., 10^{-6}) are set to manage the trade-off between accuracy and computational cost.

Dataset

Medical Pretraining Dataset. To support generalizable and clinically relevant time series modeling, we curate a large pretraining corpus spanning over 454 billion time points from various real-world healthcare settings. The dataset collection includes signals from ICUs a nd operating rooms, pediatric critical care, long-term sleep and mental health monitoring, and population-level epidemiological surveillance. All data are drawn from publicly available clinical datasets, including MIMIC-III [73], MIMIC-IV [74], PTB-XL [75], Sleep-EDF [76], and the WAVES Pediatric Waveform Database [77]. To enable the model to acquire general knowledge, we resumed training from the Time-MoE checkpoint. A full summary of included datasets and the pre-processing applied is provided in Appendix E.

Model Training

TIME-MOE: BILLION-SCALE TIME SERIES FOUNDATION MODELS WITH MIXTURE OF EXPERTS

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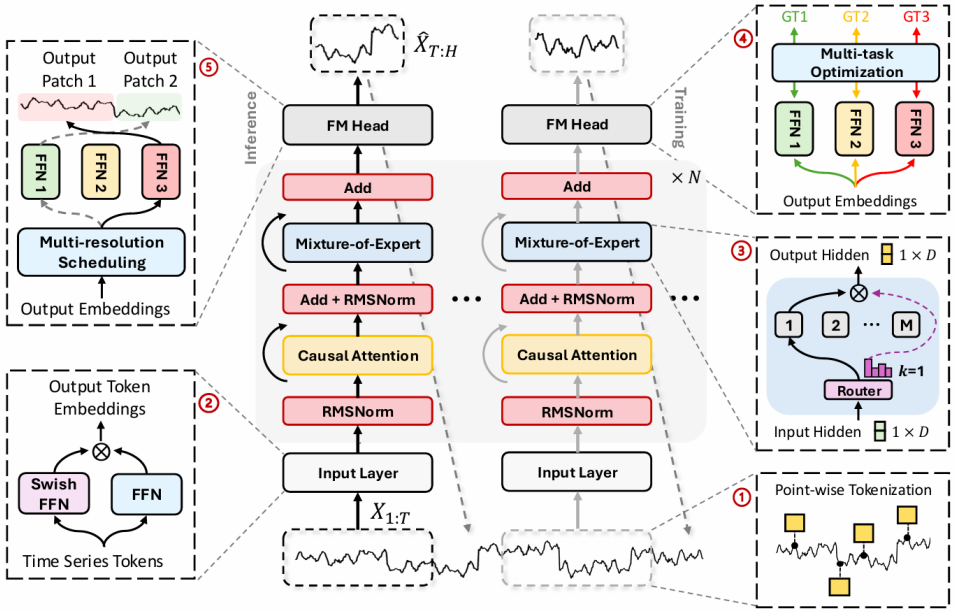

Figure 2: The architecture of TIME-MOE, which is a decoder-only model. Given an input time series of arbitrary length, ① we first tokenize it into a sequence of data points, ② which are then encoded. These tokens are processed through N-stacked backbone layers, primarily consisting of causal multi-head self-attention and ③ sparse temporal mixture-of-expert layers. During training, ④ we optimize forecasting heads at multiple resolutions. For model inference, TIME-MOE provides forecasts of flexible length by ⑤ dynamically scheduling these heads. Details about the causal multi-head self-attention are in Appendix B and illustrated in Figure 5.

Resumed training from the **Time-MoE** checkpoint

Model Training

$$\mathcal{L}_{\text{Huber}}(x, \hat{x}) = \begin{cases} \frac{1}{2}(x - \hat{x})^2, & \text{if } |x - \hat{x}| \leq \delta, \\ \delta \cdot (|x - \hat{x}| - \frac{1}{2}\delta), & \text{otherwise.} \end{cases}$$

where δ is a threshold controlling the transition between L2 and L1 loss regimes.

$$\mathcal{L}_{\text{aux}} = N \cdot \sum_{i=1}^{N} f_i \cdot r_i$$

where $f_i = \frac{1}{KT} \sum_{t=1}^{T} \mathbb{I}(\text{Expert} i \text{ selected at } t)$ and $r_i = \frac{1}{T} \sum t = 1^T s_{i,t}$ with $s_{i,t}$ being the softmax routing score.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Huber}} + \alpha \cdot \mathcal{L}_{\text{aux}}$$

Model Training

Table 1: A high-level summary of MIRA model configurations.

	Layers	Experts	K	d_{model}	d_{ff}	d_{expert}	Activated Params	Total Params
$MIRA_{small}$	8	8	2	288	1152	144	30 M	73 M
$MIRA_{base}$	12	8	2	384	1536	192	50 M	114 M
$MIRA_{large}$	12	8	2	768	3072	384	200 M	455 M

Results

Experimental Setup

Inherently Irregular Datasets:

- Exhibit irregular sampling and natural missing values

Originally Regular Datasets:

- Simulate irregularity by randomly masking 30% of time points

Evaluation Metrics. We measure the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE) as the evaluation metrics. Detailed definitions are provided in Appendix I.

Baselines

Category	Pre-training Data	Seen Target Dataset?	Evaluation Method
Full-shot	General + Medical + Target split	✓ Yes (Fine-tuned)	After supervised fine-tuning
Zero-shot (General)	General-domain (Energy, Finance, etc.)	✗ No	Direct inference / prediction
Zero-shot (Medical)	General + Medical corpus (454B)	✗ No	Direct inference / prediction

Results: In-distribution Forecasting

Objective. We evaluated in-distribution performance by holding out a portion of the pre-training datasets as test sets, ensuring no data leakage. All models were tested in zero-shot settings.

Table 3: Zero-shot forecasting performance on in-distribution datasets that are regularly sampled but contain missing values. Reported values are averaged across all prediction lengths. Lower RMSE and MAE indicate better predictions. **Red**: the best; Blue: the second best.

	Models	Ours			Baseline							
		MIRA$_{small}$	MIRA$_{base}$	MIRA$_{large}$	Moirai$_{small}$	Moirai$_{base}$	Moirai$_{large}$	Time-MoE$_{base}$	Time-MoE$_{large}$	Chronos$_{small}$	Chronos$_{base}$	Chronos$_{large}$
RMSE	SleepEDF (10^2)	0.215	0.195	0.189	0.301	0.668	0.304	0.228	0.244	0.411	0.414	0.413
	PTB-XL	0.147	0.127	0.121	0.177	0.270	0.416	0.110	0.109	0.228	0.234	0.229
	MIMIC-III	0.126	0.107	0.102	0.163	0.256	0.172	0.105	0.103	0.153	0.154	0.151
	MIMIC-IV	0.111	0.091	0.081	0.259	0.300	0.319	0.084	0.082	0.309	0.317	0.319
	WAVES	0.154	0.136	0.129	0.177	0.190	0.169	0.148	0.141	0.184	0.183	0.182
	1st Count	0	0	4	0	0	0	0	1	0	0	0
MAE	SleepEDF (10^2)	0.180	0.162	0.156	0.264	0.227	0.323	0.191	0.203	0.192	0.193	0.193
	PTB-XL	0.110	0.095	0.091	0.125	0.098	0.099	0.063	0.066	0.100	0.104	0.103
	MIMIC-III	0.106	0.089	0.084	0.141	0.164	0.138	0.081	0.078	0.079	0.080	0.080
	MIMIC-IV	0.094	0.069	0.061	0.223	0.291	0.143	0.064	0.062	0.134	0.142	0.143
	WAVES	0.129	0.112	0.106	0.157	0.181	0.155	0.124	0.116	0.119	0.118	0.117
	1st Count	0	0	3	0	0	0	1	1	0	0	0

Results: Out-of-distribution Forecasting

Table 2: Zero-shot forecasting performance on out-of-distribution datasets that are regularly sampled but contain missing values. Reported values are averaged across all prediction lengths. Lower RMSE and MAE indicate better predictions. **Red**: the best; Blue: the second best compared to zero-shot baselines; Underline: the best performance compared to full-shot baselines.

Models	Zero-shot Ours						Full-shot Time Series Models									
	MIRA_{small}		MIRA_{base}		MIRA_{large}		Contiformer		T-PatchGNN		ODE-RNN		Neural-CDE		TimesFM	
Metrics	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**
Cinc 2012 (10^1)	7.136	6.984	6.762	6.734	6.221	6.115	5.987	5.985	6.247	6.246	6.997	6.995	7.498	7.497	-	-
Heart Rate (10^{-1})	1.795	1.511	1.723	1.431	1.392	1.310	0.774	0.633	0.627	0.497	0.945	0.683	0.671	0.587	1.753	0.832
MIT-BIH	0.293	0.198	0.199	0.141	0.173	0.130	0.453	0.354	0.705	0.627	0.882	0.623	0.242	0.196	0.335	0.141
CDC-IHA (10^1)	5.976	4.684	5.729	4.502	5.534	4.401	5.211	4.103	9.522	7.974	10.068	9.052	7.892	6.766	15.633	4.408
JH COVID-19 (10^2)	0.407	0.355	0.504	0.349	0.478	0.336	0.323	0.297	0.350	0.291	0.424	0.331	0.545	0.503	2.329	0.322
ILI	1.294	1.077	1.218	1.024	1.154	1.041	0.391	0.224	0.195	0.143	0.410	0.264	0.423	0.314	2.034	1.333
1st Count	0	0	0	0	4	3	0	0	0	0	0	0	0	0	0	0

Baseline	Zero-shot Foundation Models Pre-trained on General-domain Corpora															
	Lag-Llama		TimeGPT		Timer		Moment_{small}		Moment_{base}		Moment_{large}		$\text{Moirai-MoE}_{small}$		Moirai-MoE_{base}	
	RMSE	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**
Heart Rate (10^{-1})	1.764	1.488	2.258	1.915	1.901	1.704	2.966	1.852	2.939	1.835	2.917	1.735	1.982	1.600	2.144	1.652
MIT-BIH	0.217	0.169	0.231	0.185	0.255	0.213	0.417	0.229	0.416	0.228	0.413	0.224	0.208	0.167	0.208	0.167
CDC-IHA (10^1)	6.531	4.846	6.654	4.860	6.424	4.857	17.803	5.307	17.631	5.288	17.689	5.260	7.099	5.405	7.639	5.862
JH COVID-19 (10^2)	3.596	1.432	1.879	1.580	2.647	2.328	3.077	0.553	3.097	0.554	3.064	0.549	1.452	0.725	19.391	3.190
ILI	1.780	1.366	2.011	1.077	1.882	1.492	1.570	1.056	1.566	1.054	1.566	1.052	2.001	1.678	1.983	1.664
1st Count	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Results: Out-of-distribution Forecasting

Zero-shot Foundation Models Pre-trained on General-domain Corpora

Baseline	Moirai_{small}		Moirai_{base}		Moirai_{large}		Time-MoE_{base}		Time-MoE_{large}		Chronos_{small}		Chronos_{base}		Chronos_{large}	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Heart Rate (10^{-1})	2.359	1.965	2.156	1.767	2.098	1.644	0.850	0.650	0.833	0.639	1.357	0.587	1.189	0.489	1.218	0.506
MIT-BIH	0.343	0.249	0.421	0.302	0.593	0.149	0.171	0.135	0.172	0.135	0.353	0.147	0.361	0.149	0.350	0.147
CDC-IHA (10^{1})	6.835	5.271	7.328	5.526	6.788	5.302	6.311	4.748	6.312	4.715	15.502	4.421	15.825	4.438	15.986	4.517
JH COVID-19 (10^{2})	1.917	0.695	0.991	0.474	0.614	0.402	0.596	0.402	0.512	0.371	4.826	1.031	3.835	0.551	3.478	0.521
ILI	1.995	1.671	1.871	1.561	1.808	1.499	1.288	0.951	1.366	1.015	2.054	1.400	1.940	1.308	1.870	1.252
1st Count	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Zero-shot Foundation Models Continue Pre-trained on Medical Corpora

Baseline	Moirai_{small}		Moirai_{base}		Moirai_{large}		Time-MoE_{base}		Time-MoE_{large}		Chronos_{small}		Chronos_{base}		Chronos_{large}	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Heart Rate (10^{-1})	2.047	1.685	1.907	1.536	1.601	1.263	0.648	0.524	0.603	0.488	1.049	0.555	0.965	0.488	0.902	0.458
MIT-BIH	0.239	0.204	0.274	0.190	0.219	0.166	0.201	0.155	0.185	0.143	0.311	0.137	0.320	0.139	0.306	0.127
CDC-IHA (10^{1})	6.690	5.310	6.934	5.376	6.696	5.114	6.327	4.698	6.299	4.666	14.502	4.321	14.825	4.338	14.986	4.417
JH COVID-19 (10^{2})	0.579	0.448	0.619	0.478	0.812	0.337	0.509	0.353	0.517	0.362	4.225	0.947	3.535	0.510	3.328	0.469
ILI	1.528	1.289	1.435	1.191	1.501	1.229	1.188	0.915	1.201	0.927	1.705	1.127	1.639	1.081	1.547	1.028
1st Count	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0

Results: Out-of-distribution Forecasting

1. MIRA consistently achieves **SOTA** performance
2. All model variants—continue pretrained on medical corpora—**consistently outperform** models trained on general time series data
3. MIRA **large outperforms** than **full-shot** baseline on MIT-BIH

Model Scaling and Data Behavior

Table 4: Zero-shot forecasting performance on different missing rates. Reported values are averaged across all prediction lengths. Lower RMSE and MAE indicate better predictions. **Red**: the best; Blue: the second best.

	Missing Rate	10%+		20%+		30%+		40%+		50%+		60%+		70%+		80%+		90%+		1st Count	
	Metrics (10^2)	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Ours	**MIRA**$_{small}$	3.15	2.82	3.28	2.95	3.41	3.08	3.54	3.20	3.67	3.34	3.80	3.47	3.93	3.61	4.07	3.75	4.22	3.90	0	0
	MIRA$_{base}$	2.98	2.68	3.11	2.81	3.25	2.94	3.38	3.07	3.51	3.20	3.64	3.34	3.78	3.48	3.93	3.63	4.10	3.79	0	1
	MIRA$_{large}$	2.85	2.56	2.95	2.65	3.08	2.78	3.21	2.90	3.35	3.04	3.49	3.18	3.63	3.31	3.99	3.86	3.97	3.64	8	8
General Setting	**Time-MoE**$_{base}$	3.05	2.78	3.18	2.92	3.32	3.05	3.45	3.18	3.58	3.31	3.71	3.43	3.85	3.57	3.99	3.71	4.15	3.86	0	0
	Time-MoE$_{large}$	3.20	2.91	3.33	3.04	3.47	3.17	3.60	3.29	3.73	3.41	3.86	3.54	3.99	3.67	4.13	3.81	4.28	3.96	0	0
	Moirai$_{large}$	4.80	4.18	5.39	4.64	5.68	4.55	6.02	4.83	6.09	5.06	6.18	5.02	6.31	5.09	6.51	5.24	6.71	5.36	0	0
	Moirai$_{base}$	4.84	4.15	4.93	4.10	5.20	4.23	5.64	4.53	5.78	4.61	5.80	4.60	5.95	4.97	5.95	4.73	6.06	4.87	0	0
	Moirai$_{small}$	4.91	4.22	5.04	4.20	6.12	4.93	6.23	4.98	6.44	5.49	6.47	5.18	6.59	5.60	6.66	5.20	6.86	5.71	0	0
	Chronos$_{large}$	4.71	4.10	4.85	4.22	5.02	4.31	5.31	4.52	5.58	4.70	5.82	4.91	5.97	5.10	6.15	5.32	6.43	5.49	0	0
	Chronos$_{base}$	5.02	4.40	5.21	4.51	5.43	4.68	5.62	4.81	5.85	4.93	6.02	5.12	6.21	5.31	6.40	5.49	6.67	5.62	0	0
	Chronos$_{small}$	5.32	4.71	5.49	4.82	5.68	4.93	5.81	5.02	6.01	5.20	6.21	5.32	6.40	5.49	6.58	5.61	6.83	5.72	0	0
Medical Setting	**Time-MoE**$_{large}$	2.95	2.70	3.08	2.83	3.22	2.96	3.35	3.09	3.48	3.22	3.61	3.35	3.75	3.49	3.89	3.63	4.04	3.78	1	0
	Time-MoE$_{base}$	3.05	2.78	3.18	2.91	3.32	3.04	3.45	3.17	3.58	3.30	3.71	3.43	3.85	3.57	3.99	3.71	4.15	3.86	0	0
	Moirai$_{large}$	4.41	3.76	4.92	4.20	5.35	4.41	5.69	4.76	5.93	4.80	6.09	4.92	6.23	4.98	6.48	5.29	6.62	5.30	0	0
	Moirai$_{base}$	4.55	3.92	4.78	4.07	5.18	4.26	5.49	4.46	5.78	4.64	5.70	4.51	5.87	4.97	5.92	4.75	5.96	4.71	0	0
	Moirai$_{small}$	5.02	4.21	5.36	4.40	5.55	4.98	5.97	4.92	6.25	5.34	6.43	5.11	6.44	5.35	6.57	5.41	6.58	5.30	0	0
	Chronos$_{large}$	4.30	3.89	4.58	4.02	4.83	4.20	5.02	4.31	5.32	4.51	5.61	4.73	5.83	4.91	6.02	5.10	6.31	5.32	0	0
	Chronos$_{base}$	4.61	4.10	4.83	4.20	5.02	4.31	5.32	4.51	5.58	4.70	5.82	4.91	5.97	5.10	6.15	5.32	6.43	5.49	0	0
	Chronos$_{small}$	4.83	4.31	5.02	4.43	5.32	4.51	5.58	4.70	5.82	4.91	5.97	5.10	6.15	5.32	6.43	5.49	6.71	5.61	0	0

Ablation Analysis

Table 5: Ablation studies. (**Left**) Evaluated with different model components. (**Right**) Analysis performance and inference speed across different top$_k$ setups. Lower values indicate better performance.

Component	RMSE	MAE
$MIRA_{base}$	0.154	0.118
w/o CT-RoPE	0.158	0.125
w/o MoE Block	0.157	0.122
w/o CT-Extrapolation Block	0.162	0.128

Top$_k$ Setup	RMSE	MAE	Speed (s/iter)
w/ {Top$_1$}	0.160	0.121	0.097
w/ {Top$_2$}	0.154	0.118	0.101
w/ {Top$_4$}	0.154	0.117	0.112
w/ {Top$_6$}	0.156	0.120	0.124
w/ {Top$_8$}	0.159	0.122	0.127

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